DANEBEN! SMART IOT WASTE BIN SYSTEM — CORE BILL OF MATERIALS (BOM)

Document ID: BOM-IOT-2026-CORE | Status: PRELIMINARY (BOM NOT FINAL) | Scope: 4 Major Subsystems | Date: October 2026

Pos.	Component & Technical Specification	Website / Vendor	Search Term / SKU / MPN	Qty	Est. Unit Price	Critical Remarks & Verification
01	Seeed Studio XIAO ESP32-S3 Sense Dual-Core 240MHz, 8MB PSRAM, 8MB Flash, OV2640 Cam, Mic,	Farnell (de.farnell.com) (Alt: Reichelt)	113991115 (Reichelt: XIAO ESP32S3 SEN)	1×	€14.35 (Alt: €15.00)	CRITICAL: Must be "Sense" model (MPN: 113991115). Must have 8MB PSRAM and 8MB Flash for Edge Impulse vision
02	Waterproof Ultrasonic Sensor Module JSN-SR04T / AJ-SR04M (IP67 sealed, 20–450cm, 40kHz, 2.5m	Funduino.com (Alt: Roboter-Bausatz / Amazon)	AJ-SR04M / JSN-SR04T (RBS: RBS11632)	2×	€6.21 (€12.42 for 2)	Do not cut or shorten the 2.5m coaxial cable. Plugs into onboard carrier docking bays.
03	MAX98357A I2S 3.2W Class-D Audio Amp Pure digital I2S, 3.2W @ 4Ω, 92% efficiency	Funduino.com (Alt: BerryBase.de)	MAX98357 Audioverstärker (BerryBase: ADA5770 / ADA3006)	1×	€4.08 (Alt: €4.90)	Strictly require I2S digital audio (eliminates Wi-Fi/BLE RF buzzing hum).
04	3W 4Ω / 8Ω Mini Speaker (~40 mm) Ø 40mm, 3W, 4Ω or 8Ω, -20°C to +70°C	Funduino.com (Alt: BerryBase.de) (3rd choice: Reichelt.de)	Lautsprecher 3W 4 Ohm (BerryBase: Mono-Lautsprecher 3W) (Reichelt: Ekulit AL-23P)	1×	€2.40 (Alt: €2.50 / Reichelt: €2.70)	—
Σ	**SUBTOTAL CORE ELECTRONICS (GROSS INCL. 19% VAT)**	**Farnell + Funduino**	*Master Core Subtotal*	**5×**	**€33.25** **(Alt: €38.18)**	*Subtotal for 4 core subsystems before shipping.*

**** NOTE: BoM not final — Preliminary prototype specification. Component models, vendors, and prices subject to change during laboratory testing.***